

Applications of Probability to Economics

Lecture Notes

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1 Probability concepts

In these notes we will use the following concepts from the probability theory course.

- Discrete and continuous random variables.
- Expectations and conditional expectations.
- Functions of random variables.
- Deriving probability distributions of functions of random variables.
- Variances and covariances of random variables, and their properties.

2 Allocation of a Single Object

2.1 Model of the Environment

Introduction. Consider a seller who owns one indivisible good — such as a rare painting, a concert ticket, or the right to develop a parcel of land. The seller wants to give the object away, either exchanging it for money or else. There are potential buyers, each with their own private valuation for the object. The seller does not know the valuations – and this is a key difficulty when deciding whom to give the object, whatever the objective of the seller is. This uncertainty makes probability a natural tool for analysis. In what follows we explore the economic models of three common ways, also known as mechanisms, how the seller can allocate the object. We will first consider a posted price mechanism, and then two possible auction formats. We begin with a mathematical description of the environment, and see how probability theory is used to model the uncertainty in this situation.

Valuations and payoffs. There are $n \geq 1$ buyers. Each buyer i has a *valuation* v_i for the object, measured in money. We assume that if buyer i receives the object and pays t_i , her utility is

$$u_i = v_i - t_i,$$

and if she does not receive the object (and pays nothing), $u_i = 0$.¹ Hence a buyer is willing to pay a price p iff $v_i \geq p$. In this sense, v_i is the buyer's *maximum willingness to pay* for the object. When the object is allocated to buyer i , a surplus of v_i is created. Transfers t_i simply redistribute this surplus between buyers and the seller; they do not change the total value created by allocating the object to a buyer.

¹In economics this form of utility is known as quasi-linear (linear in money).

Seller’s uncertainty and random valuations. Each buyer knows their own valuation, but not the valuations of others, and moreover the seller does not know values of buyers. To model this uncertainty, we treat buyers’ valuations as random variables

$$V_i \sim F \quad (i = 1, \dots, n),$$

drawn independently from a common cumulative distribution function F . The seller knows the distribution F but not the realizations $(v_1, \dots, v_n)$, while each buyer i knows the realization v_i . Hence, we assume that valuations are *independent and identically distributed (i.i.d.)* random variables. Independence of valuations means that knowing one buyer’s value conveys no information about others. Identical distribution means that the seller views buyers as completely symmetric.²

Together, these assumptions define the simplest probabilistic setting for studying allocation mechanisms: one seller and one object, many symmetric buyers with private values, and the sellers’ uncertainty about the values of the buyers. For concreteness we will also assume a specific distribution of valuations F – we will use the simplest continuous distribution available, the uniform. Specifically, we let:

$$V_i \sim U[0, 1], \quad (i = 1, \dots, n)$$

In the problem sets and exams you will be asked to work with other distributions too. So, while the uniform simplifies the computations, pay attention to the general way we derive results which is applicable to any other continuous distribution.

Seller’s objective. The seller’s goal determines what kind of allocation rule will be optimal. We will broadly keep in mind two objectives.

One possible objective is **efficiency**, or welfare maximization. An efficient allocation gives the object to the person who values it most, maximizing total surplus created. This is the perspective of a planner or policymaker who wants resources to end up where they create the most value. The transfers that such seller can collect along the way can be, for example, redistributed among the buyers, and serve only as a mechanism to allocate the object, not the goal in itself.

A different objective is **revenue maximization**. Here the seller is self-interested and cares only about the expected payment they receive. For simplicity we will assume that the seller herself has no value for the object, so revenue maximization is the same as profit maximization in this case.

These two objectives — efficiency and revenue — do not always coincide. The efficient

²In economics these assumptions are known as private values case.

allocation rule may generate less revenue for the seller, while the revenue-maximizing rule may lead to inefficiency. Much of economic theory is about understanding and quantifying this trade-off.

No uncertainty benchmark. Suppose there is *no uncertainty*: all valuations are known to buyers and the seller. Then to maximize welfare, the seller simply allocates the object to the highest-value buyer^{footnote}Here $\arg \max_i v_i$ is the argument i that makes v_i as high as possible.

$$i^* = \arg \max_i v_i;$$

Similarly, if the seller wants to maximize revenue, she can make a take-it-or-leave-it offer to i^* at price $p = v_{i^*} - \varepsilon$, where $\varepsilon > 0$ is arbitrarily small. Buyer i^* accepts and the seller captures (almost) the entire surplus. More generally, with full information and commitment, transfers can be set to implement virtually any reasonable objective.

Hence, the really interesting case—and the need to use probability—arises under uncertainty: the seller does not observe valuations, so mechanisms must balance information and incentives.

2.2 Posted Price Mechanism

While there are many ways to allocate an object, the most familiar and straightforward is to post a price and let buyers decide whether to buy. This is what supermarkets do. We analyze this *posted price* mechanism first. Our goal is to derive the optimal price and the corresponding revenue/welfare.

Rules of the posted-price mechanism.

- The seller chooses and announces a price $p \in [0, 1]$.³
- Each buyer i decides whether to buy: buyer i is willing to purchase the object if and only if $V_i \geq p$, where V_i is their valuation.
- If at least one buyer is willing to buy, one of them is selected uniformly at random from the set of willing buyers. The selected buyer pays the posted price p and receives the object.
- If no buyer is willing to pay p , the object remains unsold and no trade occurs.

³For $p > 1$, demand is zero under the uniform distribution, so we restrict attention to the interval $[0, 1]$.

Expected Revenue. In a posted price mechanism, the seller either receives p if at least one buyer wants to buy, or zero if no one is willing to buy. So, the revenue that the seller receives is a discrete random variable. What does it mean to maximize revenue in this context? While there are many ways one can approach this question, a standard approach used in economics employs the notion of expectation of a random variable. We assume that the seller is trying to choose such a price p which results in the highest expected revenue. In our case the expected revenue is

$$\Pi(p) = p \cdot \Pr(\text{sale at price } p) = p \Pr(\exists i \text{ s.t. } V_i \geq p).$$

Using the complement probability, independence, and assumption of uniform distribution of valuations, we get

$$\Pr(\exists i \text{ s.t. } V_i \geq p) = 1 - \Pr(V_1 < p, \dots, V_n < p) = 1 - p^n,$$

so the expected revenue is

$$p(1 - p^n),$$

as depicted in Figure 1.

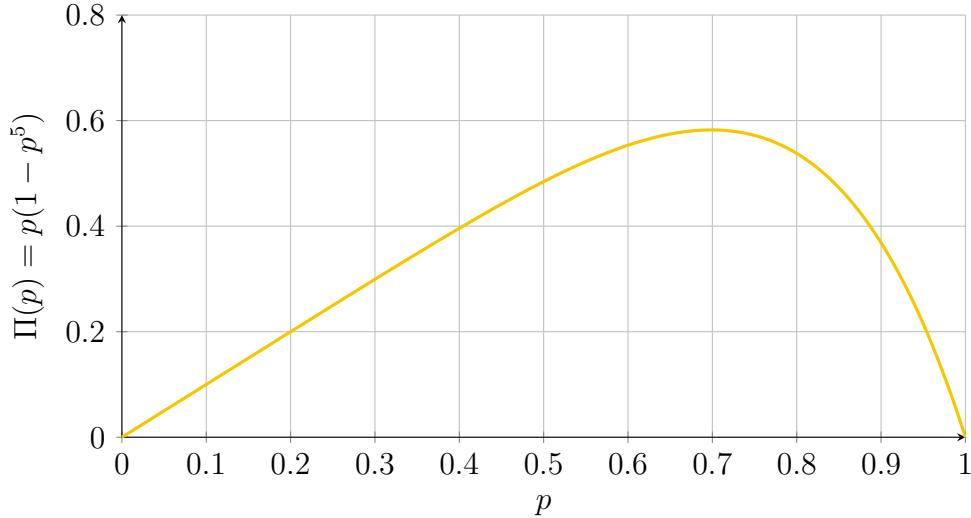


Figure 1: Expected revenue $p(1 - p^n)$ for $n = 5$ as a function of price.

Optimal Price. The seller chooses p to maximize expected revenue:

$$\max_{p \in [0,1]} p(1 - p^n).$$

First-order condition:

$$1 - (n + 1)p^n = 0 \implies p^{rev} = (n + 1)^{-1/n}.$$

To make sure that we have indeed found the maximum of the expected revenue, we need to check second-order condition. Second derivative is:

$$-n(n+1)p^{n-1} < 0 \text{ for } p > 0,$$

and $\Pi(0) = \Pi(1) = 0$, so p^{rev} is the global maximizer. Thus optimal price and the corresponding expected revenue are

$$p^{rev} = (n+1)^{-1/n}, \quad \Pi^{rev} = \frac{n}{n+1} (n+1)^{-1/n}.$$

Efficiency note. This allocation can be inefficient. If multiple buyers are willing to buy, the object is assigned randomly and may not go to the highest-value buyer; on the other hand if all valuations fall below p , which can happen with positive probability, there is no trade even though allocating an object still generates positive surplus.

Welfare-maximizing posted price mechanism. In the previous part, we examined the posted-price mechanism from the perspective of *revenue maximization*. We now turn to a different objective: choosing a posted price to maximize *welfare*. A natural question arises: who sets a higher price — a welfare-maximizing seller or a revenue-maximizing seller?

2.2.1 Advanced welfare analysis

Expected surplus. Given a price p , only buyers with $V_i \geq p$ are willing to buy. Hence, the price acts as a *filter*: only buyers with sufficiently high valuations are allowed to purchase the good. For example, setting $p = 0$ means there is no filtering and the object is allocated randomly among all buyers, whereas setting $p = 0.5$ means that only buyers whose valuations exceed 0.5 are eligible. In this way, the price becomes a simple but potentially powerful tool for improving the allocation when welfare is the objective.

Let S denote the surplus generated by an allocation at price p . If the good is sold to buyer i , then $S = V_i$. The surplus S is a random variable because:

1. the set of buyers willing to buy depends on their random valuations, and
2. the buyer selected from this set is also random.

We are interested in the expected surplus $\mathbb{E}[S]$.

If the buyer who accepts price p is chosen randomly from those willing to buy, the only

information we have about this buyer is that $V_i \geq p$. Therefore,

$$\mathbb{E}[S] = \mathbb{E}[V_i \mid V_i \geq p] \Pr(\exists i \text{ such that } V_i \geq p).$$

To compute this expression, we first derive the conditional pdf of V_i given that $V_i \geq p$, and then find the corresponding conditional expectation.

Conditional density and expectation. Let $V \sim \text{U}[0, 1]$ with pdf $f(v) = 1$ for $v \in [0, 1]$ and cdf $F(v) = v$. Fix a posted price $p \in [0, 1]$. Consider the event $A = \{V \geq p\}$ with probability

$$\Pr(A) = \Pr(V \geq p) = 1 - F(p) = 1 - p.$$

The conditional cdf of V given A is

$$F_{V|A}(v) = \Pr(V \leq v \mid V \geq p) = \begin{cases} 0, & v < p, \\ \frac{F(v) - F(p)}{1 - F(p)} = \frac{v - p}{1 - p}, & p \leq v \leq 1, \\ 1, & v > 1. \end{cases}$$

Differentiating on $(p, 1)$ yields the conditional pdf:

$$f_{V|A}(v) = \frac{f(v)}{1 - F(p)} = \frac{1}{1 - p}, \quad v \in [p, 1],$$

and $f_{V|A}(v) = 0$ otherwise. Hence, $V \mid \{V \geq p\} \sim \text{U}[p, 1]$.

The corresponding conditional expectation is

$$\mathbb{E}[V \mid V \geq p] = \int_p^1 v f_{V|A}(v) dv = \frac{1}{1 - p} \int_p^1 v dv = \frac{1}{1 - p} \cdot \frac{1 - p^2}{2} = \frac{1 + p}{2}.$$

Interpretation. Conditioning on $V \geq p$ truncates the distribution to $[p, 1]$ and re-normalizes its mass. For the uniform case, the conditional mean is simply the midpoint of that interval, $(1 + p)/2$. When $p = 1$, the probability of sale is zero and no trade occurs.

Expected surplus and the surplus-maximizing price. With n i.i.d. buyers $V_i \sim \text{U}[0, 1]$, a sale occurs if and only if at least one buyer's valuation exceeds p :

$$\Pr(\text{sale at } p) = 1 - \Pr(V_1 < p, \dots, V_n < p) = 1 - p^n.$$

Conditional on sale, the buyer is selected uniformly at random among those willing to buy, and hence their valuation follows $V \mid (V \geq p) \sim U[p, 1]$, with

$$\mathbb{E}[V \mid V \geq p] = \frac{1+p}{2}.$$

Note that this conditional mean does not depend on n —only the probability of sale does.

Assuming zero production cost, total surplus from trade equals the buyer's valuation. Thus, the *expected surplus* under posted price p is

$$\Pr(\text{sale at } p) \cdot \mathbb{E}[V \mid V \geq p] = (1 - p^n) \frac{1+p}{2}, \quad p \in [0, 1].$$

The welfare-maximizing posted price solves

$$\max_{p \in [0, 1]} \max_{p \in [0, 1]} \frac{(1 - p^n)(1 + p)}{2}.$$

For an interior optimum, the first-order condition is

$$\frac{1}{2} \left[(1 - p^n) - np^{n-1}(1 + p) \right] = 0 \iff 1 - p^n = np^{n-1}(1 + p).$$

This equation characterizes the optimal welfare-maximizing price p^{sur} (with $p^{\text{sur}} = 0$ for $n = 1$, and $p^{\text{sur}} > 0$ for $n \geq 2$; for example, $p^{\text{sur}} = 1/3$ when $n = 2$).

The trade-off is transparent: increasing p raises the conditional value $(1 + p)/2$ of the buyer who trades, but lowers the probability of sale $1 - p^n$. The optimal p^{sur} balances these two effects.

Welfare-maximizing vs. revenue-maximizing prices. The revenue-maximizing price p^{rev} solves

$$1 - (n + 1)p^n = 0,$$

while the welfare-maximizing price p^{sur} satisfies

$$1 - p^n = np^{n-1}(1 + p).$$

We can rewrite the former as:

$$1 - (n + 1)p^n = np^{n-1}.$$

Because the right-hand side of both equations is a decreasing function of p , it follows that

$$p^{\text{sur}} < p^{\text{rev}}.$$

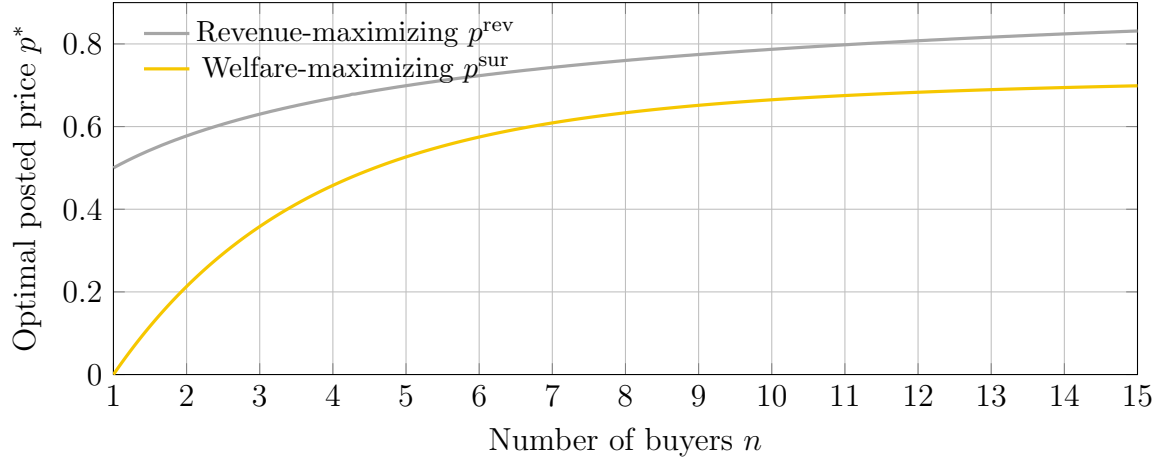


Figure 2: Welfare-maximizing (gold) and revenue-maximizing (gray) posted prices as functions of the number of buyers n . As n increases, both prices rise, but the welfare-maximizing price remains strictly below the revenue-maximizing one for all n . The gap reflects the welfare seller’s preference for a higher probability of trade rather than extracting maximum revenue per sale.

We depict both prices as functions of n in Figure 2.

Checks. For $n = 1$, $p^{\text{rev}} = 1/2$, while $S(p) = (1 - p^2)/2$ is maximized at $p^{\text{sur}} = 0$. For $n = 2$, $p^{\text{rev}} = (1/3)^{1/2} \approx 0.577$, whereas the first-order condition for S yields $p^{\text{sur}} = 1/3 \approx 0.333$. Economically, raising p increases the conditional value of the trading buyer but lowers the probability of trade; a welfare-maximizing seller favors more frequent trades and therefore sets a lower price than a revenue-maximizing one.

2.3 Second-Price Auction

Another common way to allocate an object is the **English (ascending)** auction where the price starts at zero and raises until only one bidder remains willing to pay this price. This is what you might be familiar with from movies. Famous real life example is Tokyo’s Toyosu fish market, pictured in Figure 3, that sells high-value tuna via such auction: the price is called upward, bidders drop out as it rises, and the last bidder wins at the standing price. The important observation is that the winner pays roughly the price at which the *second-to-last* bidder dropped out. Interestingly, under i.i.d. private values, this dynamic format is outcome-equivalent to another, somewhat peculiar, auction format – the sealed-bid **second-price auction**.

Rules of the second-price auction.

- Each bidder simultaneously submits one bid in a sealed envelope.



Figure 3: Tokyo Toyosu Tuna Auction

- The auctioneer chooses the highest bidder as a winner (ties broken at random).
- The winner pays the *second-highest* bid and gets the object.

Strategic behavior. Behavior of bidders in such an auction has a very convenient property. Fix other bids. If you bid *above* your value, you may win and pay more than your value (negative surplus). If you bid *below* your value, you may lose when the price would have been below your value (foregoing positive surplus). Bidding your true value avoids both problems. Thus, each bidder will bid his or her true value:

$$\text{bid of } i = v_i,$$

and this is what we assume will happen in such an auction. Note that the resulting allocation is always efficient: the highest-value bidder wins. Also note how charging second-highest price effectively induces bidders to reveal their true value in their bid. This is in contrast to the sealed-bid first-price auction that we will see next.

Expected revenue. When a seller uses an auction, there is nothing else the seller has to choose (there is no price like in the posted price mechanism). For simplicity, suppose there are only two bidders $n = 2$. Then when each bidder truthfully bids his or her own value, the seller's revenue is given by the second-highest value, which is the minimum of the two values:

$$V_{(2)} = \min\{V_1, V_2\}.$$

This second-highest value $V_{(2)}$ is a function of two random variables and hence is itself a random variable. As before, we focus on its expected value. To compute the expected value we first derive its cdf:

$$F_{V_{(2)}}(m) = \Pr(V_{(2)} \leq m) = 1 - \Pr(V_1 > m, V_2 > m).$$

Note how we use the complement probability above – the complement of the event that the minimum of the two is lower than m is the event that both values are higher than m . Using independence assumption together with the uniform distribution of valuations we get

$$F_{V_{(2)}}(m) = 1 - (1 - m)^2,$$

for $0 \leq m \leq 1$. So, now differentiating cdf we can get pdf $f_{V_{(2)}}(m) = 2(1 - m)$, and directly compute the expectation:

$$\mathbb{E}[V_{(2)}] = \int_0^1 m 2(1 - m) dm = \left[m^2 - \frac{2}{3}m^3 \right]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}.$$

Thus, the expected revenue of the seller with two bidders is $1/3$.

General n . For general number of bidders n and uniform distribution of valuations it can be demonstrated that:

$$\mathbb{E}[\text{Revenue in SPA}] = \frac{n - 1}{n + 1}.$$

Note that this simplifies to $1/3$ for $n = 2$ as we derived above. However, we do not prove this general result here.

2.4 First-Price Auction

Another common selling mechanism is **Dutch (descending)** auction which starts from a high price that falls until a bidder accepts; the first to accept wins and pays the standing price. The Dutch flower auctions use this format: a price clock depicted in Figure 4 starts high and ticks down until the first buyer accepts. Under i.i.d. private values, this dynamic format is outcome-equivalent to the sealed-bid **first-price auction**.

Rules of the first-price auction.

- Each bidder simultaneously submits one bid in a sealed envelope.
- The auctioneer chooses the highest bidder as a winner (ties broken at random).



Figure 4: Price clocks at Dutch flower auction.

- The winner pays *their own* bid and gets the object.

Strategic behavior. In a first-price auction you pay your own bid, so bidding your full value is too aggressive: if you win, your surplus would be zero. Each bidder therefore *shades*—chooses $b < v$ —to trade off a higher winning probability (from bidding higher) against a lower surplus if they win (since they pay that bid). It turns out that under the uniform distribution of valuations and our assumptions, in equilibrium each bidder with valuation v will make a bid:

$$\text{bid } i = \frac{n-1}{n} v,$$

so with two bidders $b(v) = \frac{1}{2}v$, with three bidders $b(v) = \frac{2}{3}v$, etc. You will learn to derive this result if you take an advanced game theory course, but for now we will simply assume that bidders follow this strategy.

Expected revenue. With two bidders, the seller receives half of the maximum of the two values. This is again a random variable, and again we will be interested in its expectation. As before Let $V_{(1)} = \max\{V_1, V_2\}$. We shall derive cdf and then pdf of this

random variable. For $x \in [0, 1]$,

$$F_{V_{(1)}}(x) = \Pr(V_{(1)} \leq x) = \Pr(V_1 \leq x, V_2 \leq x) = x^2, \quad f_{V_{(1)}}(x) = 2x,$$

so

$$\mathbb{E}[V_{(1)}] = \int_0^1 x \cdot 2x \, dx = \frac{2}{3}.$$

Therefore expected revenue (expected winning bid) is

$$\mathbb{E}[\text{Revenue in FPA}] = \frac{1}{2} \mathbb{E}[V_{(1)}] = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

Remarkably this is the same expected revenue we derived for SPA. It is not a coincidence, and is true much more generally. This observation is a special case of the *revenue equivalence theorem* which you might encounter in advanced microeconomic courses. However, here we do not explore this result further.

Comparing Auctions vs. Posted Price. For general n i.i.d. uniform bidders, revenue equivalence gives

$$\mathbb{E}[\text{Revenue FPA}] = \mathbb{E}[\text{Revenue SPA}] = \mathbb{E}[V_{(n-1)}] = \frac{n-1}{n+1},$$

while the optimal posted price revenue is

$$\Pi^* = \frac{n}{n+1} (n+1)^{-1/n}.$$

Figure 5 compares the two.

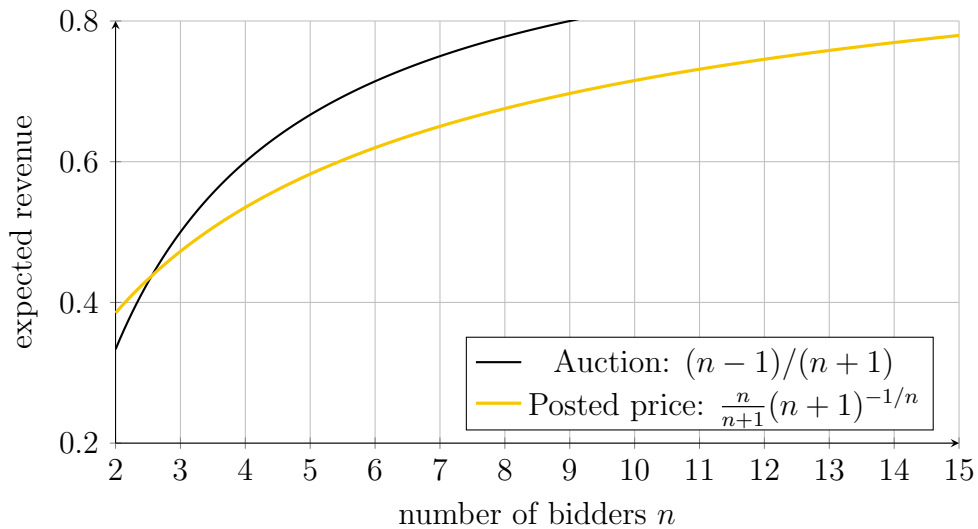


Figure 5: Expected revenue vs. number of bidders ($U[0, 1]$). Auctions weakly dominate for $n \geq 3$.

We conclude with a couple of observations for our specific simple environment.

- Using auctions result in efficient allocation – the object goes to the buyer with a highest valuation, while a posted price mechanism can be inefficient.
- Using auctions also generates higher revenue than an optimal posted price mechanism for $n \geq 3$ buyers.

3 Market for Lemons: Adverse Selection

4 Newsvendor model: Daily Cinnabons

For our next application, consider a cafe that sells freshly baked cinnabons that spoil at day's end. If the cafe *overstocks*, unsold pastries are wasted. If it *understocks*, some customers who would have bought leave empty-handed. The uncertainty here appears due to the uncertainty in how many buyers will want to buy. The decision how much of the cinnabons to stock must trade-off between the risk of leftovers and the risk of stockouts. Probability is the tool that helps to understand this tradeoff and come up with a precise objective we can optimize.

Decision and uncertainty. The cafe chooses how much cinnabons to stock, that is a *stock level* $q \geq 0$ each morning. The per-unit selling price is p (fixed for the day), and the wholesale cost is c per unit. Demand during the day is uncertain; we model demand D as a nonnegative continuous random variable with cumulative distribution function (cdf) F and density f .

Notice that all quantities are assumed to be continuous, even though in reality items such as cinnamon rolls come in discrete units. This is a standard simplification that makes the analysis more tractable without sacrificing any meaningful economic insight. Moreover, discrete demand models can also be used—and in fact, they can be microfounded from the valuation model introduced in the auction section—but the continuous formulation yields more compact expressions while preserving identical economic intuition.

Expected profit. Profit maximization is a standard objective in such problems. When cinnabons unsold during the day are worthless (in other words salvage value is zero), realized profit is

$$\pi(q; D) = p \cdot \min\{D, q\} - cq.$$

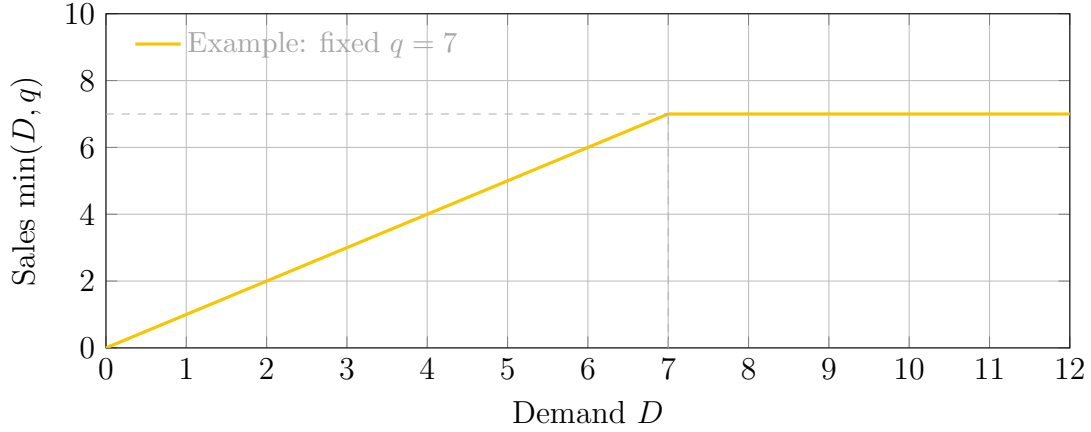


Figure 6: The function $f(D) = \min(D, q)$ for a fixed stock level q . Sales rise linearly with demand while $D < q$, but remain capped at q once inventory runs out.

The min function appears because the cafe cannot sell more cinnabons than it stocks. This profit is a function of a random variable D , and hence itself is a random variable. As before we assume that the cafe chooses its stock level q to maximize expected profit:

$$\max_{q \geq 0} \mathbb{E}[p \min(D, q) - cq].$$

Using the linearity of expectations we get the expression for expected profit as a function of stock level $\Pi(q)$:

$$\Pi(q) = p\mathbb{E}[\min(D, q)] - cq.$$

The technical difficulty is to figure out the exact form the expectation in the above expression takes. For this notice that the min function, depicted in Figure 6, actually consists of two parts stitched together, a part where sales are exactly equal to demand, and a part where sales are equal to stock level q . Specifically, using LOTUS we get

$$\mathbb{E}[\min(D, q)] = \int_0^q x f(x) dx + \int_q^\infty q f(x) dx.$$

This can be further simplified:

$$\mathbb{E}[\min(D, q)] = \int_0^q x f(x) dx + q[1 - F(q)].$$

Hence,

$$\Pi(q) = p\left(\int_0^q x f(x) dx + q[1 - F(q)]\right) - cq.$$

Now as we know the exact shape of $\Pi(q)$ we can take the first-order condition (FOC) to

find an interior optimum $q^* > 0$:

$$p[1 - F(q^*)] = c \iff F(q^*) = 1 - \frac{c}{p} = \frac{p - c}{p}.$$

Since $\Pi''(q) = -pf(q) \leq 0$, the solution is unique when $f > 0$ around q^* .

Interpretation. Choose the stock level q^* so that the probability of a stockout, $\Pr(D > q^*) = 1 - F(q^*)$, equals the *cost ratio* c/p . Equivalently, $F(q^*) = (p - c)/p$ balances the marginal *overage* cost (c per extra unit) with the marginal *underage* benefit (p per unit sold). This is the classic critical-fractile rule with zero salvage value.⁴

Example: uniform demand. For concreteness we apply the above optimality condition to a specific distribution of demand. If $D \sim U[0, 1]$, then $F(d) = d$ on $[0, 1]$, so

$$q^* = F^{-1}\left(\frac{p - c}{p}\right) = \frac{p - c}{p}.$$

The rule is transparent: stock a fraction $(p - c)/p$ of the maximum possible demand. For example, if $p = 5$ and $c = 2$, then $q^* = 3/5 = 0.6$ of the demand range (i.e., 60% of the day's maximum possible demand).

5 Diversification and Portfolio Theory

One of the oldest pieces of investment advice is simple and wise: “*Don’t put all your eggs in one basket.*” This proverb captures the essence of **diversification** — the idea that spreading your investments across several assets can reduce risk without necessarily lowering expected return.

Probability theory allows us to move beyond this common-sense statement. It gives us a precise, **quantitative language** for describing uncertainty and a set of **tools** for turning intuitive investment rules into mathematical criteria. Instead of merely saying that diversification is “good,” probability lets us measure *how much* it reduces risk and *by how much* it can smooth out returns.

Simple Example. Consider an investor who can choose among three independent assets.⁵ Probability comes in while modeling the uncertain asset returns. Specifically, we

⁴With salvage $s \in [0, c]$, the FOC becomes $F(q^*) = \frac{p-c}{p-s}$.

⁵We define assets as stocks, bonds, cash, cryptocurrency, real estate, and any other property acquired today to generate future returns, whether through capital gains upon sale (appreciation of price) or periodic payments such as dividends.

assume that each asset provides a return captured by a random variable U_i , for $i = 1, 2, 3$, where

$$U_i \sim U[0.5, 1.5].$$

That is, each asset's return fluctuates randomly between 0.5 and 1.5, representing uncertainty about performance. If the investor allocates an amount x_i to asset i , then their payoff from this asset is

$$x_i U_i.$$

So, the total portfolio return $U(\alpha)$ is simply

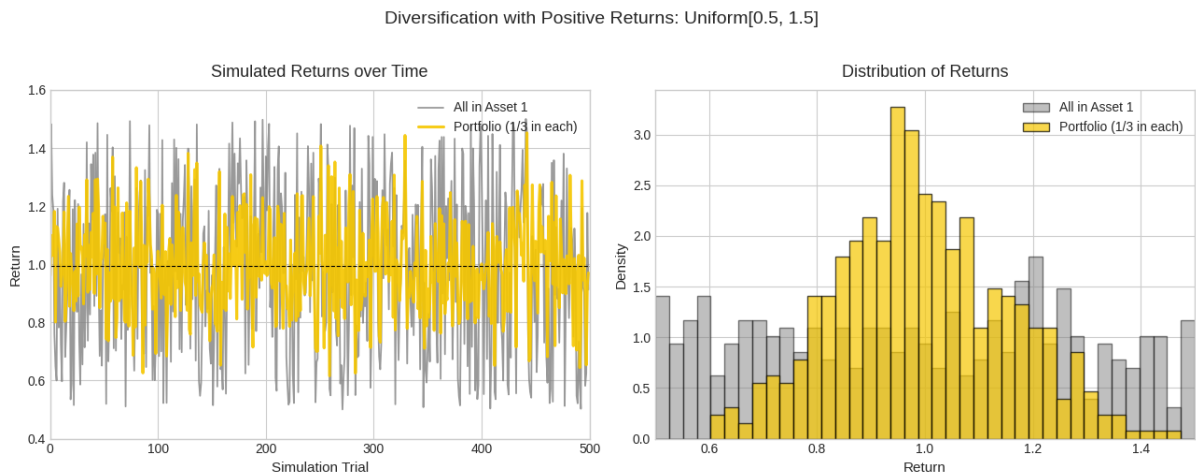
$$U(\alpha) = \frac{x_1 U_1 + x_2 U_2 + x_3 U_3}{x_1 + x_2 + x_3} = \alpha_1 U_1 + \alpha_2 U_2 + \alpha_3 U_3,$$

where $\alpha_i = x_i / \sum x_j$ is a share of total wealth invested in asset i . Hence, the total portfolio return is given by some weighted average of the returns of individual assets in the portfolio. Note that since each U_i is a random variable, the total portfolio return is also random.

Simulation. For illustration we will compare two investment strategies:

1. **Undiversified portfolio:** invest everything in asset 1 ($\alpha_1 = 1$).
2. **Diversified portfolio:** invest equally in all three assets ($\alpha_i = 1/3$ for all i).

The following graphs show a simulation of 500 independent investment outcomes for both portfolios. The left panel plots the returns over time (simulation trials), and the right panel shows the distribution (histogram) of returns.



We can immediately observe two features:

1. **Dampened fluctuations:** The yellow line (diversified portfolio) moves more smoothly around its mean than the grey line (single asset). Random highs and lows from different assets tend to offset one another, reducing volatility.
2. **Concentrated distribution:** The histogram of the diversified portfolio is much more concentrated around the mean value of 1. The expected return stays constant, but the risk, measured by the spread of possible outcomes, clearly decreases.

Why does this happen? When returns of individual assets are *independent*, their variations average out when combined. This simple example captures the key insight of modern **portfolio theory**: diversification *smooths* the distribution of possible outcomes. Probability provides the formal framework to quantify this effect, to compute expectations, variances, and correlations, and ultimately to design portfolios that balance risk and return systematically rather than intuitively. Next we explore this in detail.

Model Setup. For simplicity we will work with two assets only. Let R_1 and R_2 be random variables denoting the returns of two assets. Let $\alpha \in [0, 1]$ be the portfolio weight on asset 1, so the weight on asset 2 is $1 - \alpha$. The *portfolio return* is the convex combination

$$R_p(\alpha) = \alpha R_1 + (1 - \alpha) R_2.$$

Define the means, variances, and covariance:

$$\mu_i = \mathbb{E}[R_i], \quad \sigma_i^2 = \text{Var}(R_i), \quad \sigma_{12} = \text{Cov}(R_1, R_2).$$

Expected Return and Risk. As is standard in finance we shall be interested only in the expected return and risk of the portfolio as captured by the variance of its return. Remember that this is a simplification, which, however, simplifies the analysis a lot. By linearity of expectation, the portfolio's expected return denoted by $\mu_p(\alpha)$ is linear in the weights:

$$\mu_p(\alpha) = \mathbb{E}[R_p(\alpha)] = \alpha \mu_1 + (1 - \alpha) \mu_2.$$

Risk is commonly measured by variance:

$$\sigma_p^2(\alpha) = \text{Var}(R_p(\alpha)) = \alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2 + 2\alpha(1 - \alpha) \sigma_{12}.$$

Recall that we have derived the above expression as a property of variance. Unlike μ_p , the function $\sigma_p^2(\alpha)$ is *nonlinear* in α and depends on the *covariance* σ_{12} .

When assets move together (positive covariance), shocks reinforce each other, so diversification helps less. When assets move in opposite directions (negative covariance), shocks

partially cancel, and the portfolio's variance can drop dramatically. In fact, with sufficiently negative correlation, you can reduce risk more than by simply picking the less volatile asset in isolation. We shall consider two simple objectives first.

Simulation Example We now return to the simulated example from the beginning of this section. For any single asset U_i , the expected value and variance are

$$\mathbb{E}[U_i] = \frac{0.5 + 1.5}{2} = 1, \quad \text{Var}(U_i) = \frac{(1.5 - 0.5)^2}{12} = \frac{1}{12}.$$

- **Undiversified portfolio.** Let $R^{(1)} = U_1$ be revenue of this portfolio. Then:

$$\mathbb{E}[R^{(1)}] = 1, \quad \text{Var}(R^{(1)}) = \frac{1}{12}, \quad \sigma(R^{(1)}) \approx 0.2887.$$

- **Equal-weight diversified portfolio.** Let

$$R^{(3)} = \frac{1}{3}(U_1 + U_2 + U_3).$$

be revenue from diversified portfolio. Then

$$\mathbb{E}[R^{(3)}] = \frac{1}{3}(\mathbb{E}U_1 + \mathbb{E}U_2 + \mathbb{E}U_3) = 1,$$

$$\text{Var}(R^{(3)}) = \frac{1}{9}(\text{Var}(U_1) + \text{Var}(U_2) + \text{Var}(U_3)) = \frac{1}{108}.$$

Maximize expected return. Since $\mu_p(\alpha) = \alpha\mu_1 + (1 - \alpha)\mu_2$ is linear in α , the maximizer is at a boundary:

$$\alpha^* = \begin{cases} 1, & \text{if } \mu_1 > \mu_2, \\ 0, & \text{if } \mu_2 > \mu_1, \\ \text{any } \alpha \in [0, 1], & \text{if } \mu_1 = \mu_2. \end{cases}$$

Investment advice here is simple – if one cares only about expected return and not the risk then simply invest all in the asset with the highest expected return.

Minimize risk. Minimize $\sigma_p^2(\alpha)$ over α :

$$\min_{\alpha} \alpha^2\sigma_1^2 + (1 - \alpha)^2\sigma_2^2 + 2\alpha(1 - \alpha)\sigma_{12}.$$

First order condition is:

$$\frac{d\sigma_p^2}{d\alpha} = 2\alpha\sigma_1^2 - 2(1 - \alpha)\sigma_2^2 + 2(1 - 2\alpha)\sigma_{12} = 0.$$

Solve for α :

$$\alpha_{\min} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

provided the denominator > 0 (which holds unless R_1 and R_2 are perfectly comoving and identical up to scale). If $\alpha_{\min} < 0$, it means the variance-minimizing strategy involves a *negative position* in asset 1. In financial terms, this corresponds to **short selling** asset 1 (borrowing and selling it), while investing more than the total wealth in asset 2 (a leveraged long position). Such a portfolio uses asset 1 as a **hedge**: its returns move in the opposite direction to asset 2, so a negative weight offsets part of the second asset's volatility. In practice, when short sales are not allowed, we restrict weights to $[0, 1]$, and a negative $\alpha_{\min}$ simply implies that the unconstrained optimum lies outside the feasible region. The constrained optimum then occurs at the boundary:

$$\alpha^* = 0,$$

meaning the investor holds only asset 2.

The corresponding minimum variance is

$$\sigma_{\min}^2 = \frac{\sigma_1^2\sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}.$$

A subtle but important lesson here is that even when $\mu_1 = \mu_2$ (same expected return) and $\sigma_1 > \sigma_2$ (asset 1 is riskier), the variance-minimizing portfolio *need not* be the corner $\alpha = 0$ (all in the safer asset). If correlation is not too high, mixing in *some* of the riskier asset can *reduce* overall variance thanks to imperfect comovement.

Mean–Variance curve. The set of all achievable (mean, variance) pairs by varying $\alpha \in [0, 1]$ forms a curve in the mean–variance plane. Figure 7 gives a concrete plot with parameters chosen for illustration. When $\alpha = 1$ we are in the leftmost end of the curve, when $\alpha = 0$ we are at the rightmost end. For intermediate values we are somewhere on the yellow curve.

Mean–Variance Investor. Now as we have all the building blocks we can complete the picture by introducing the classical problem of a mean–variance investor. This is a simplified model of portfolio choice where an investor is assumed to care only about expected return and risk of the portfolio. Specifically, we assume that an investor has

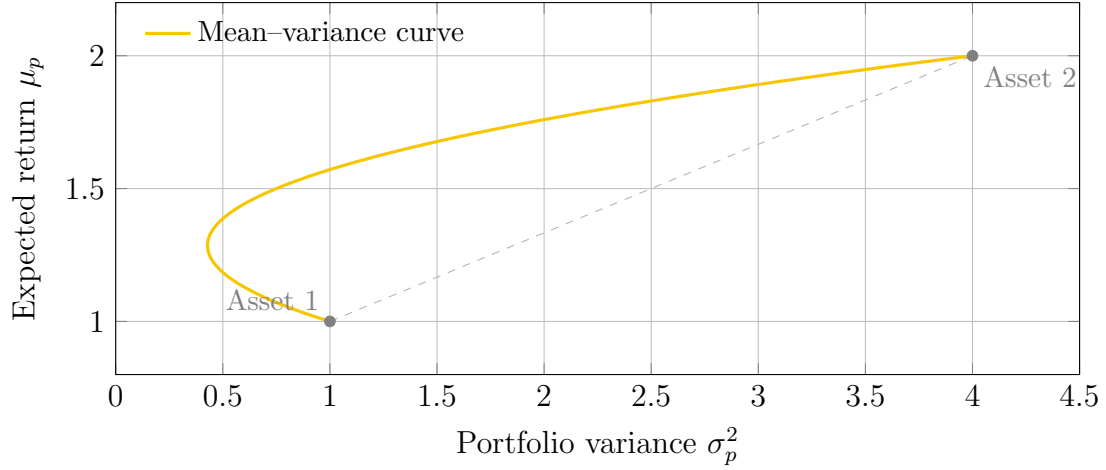


Figure 7: Risk–return combinations for two assets with $\mu_1 = 1$, $\mu_2 = 2$, $\sigma_1 = 1$, $\sigma_2 = 2$, and $\text{Cov}(R_1, R_2) = -1$. The yellow curve shows all achievable portfolios.

the following linear utility function that depends only on these two parameters:

$$U(\mu, \sigma^2) = \mu - \frac{\lambda}{2} \sigma^2, \quad \lambda > 0,$$

where μ is expected return, σ^2 is variance (risk), and λ is the coefficient of risk aversion. An *indifference curve* $U(\mu, \sigma^2) = \bar{U}$ is therefore an upward-sloping line in the (σ^2, μ) plane:

$$\mu = \bar{U} + \frac{\lambda}{2} \sigma^2,$$

with slope $\frac{\lambda}{2}$: higher risk requires higher mean to keep utility unchanged.

We assume that there are two available assets. For these given assets, the set of all achievable portfolios traces a *mean–variance curve* (the “feasible set”) in the (σ^2, μ) plane. This plays the role of a *budget constraint*: the investor cannot choose (μ, σ^2) arbitrarily but only points attainable from the assets. The optimal portfolio is found where the *steepest* attainable indifference line is *tangent* to the feasible set.

Then the problem of the investor is

$$\max_{\alpha} \mu_p(\alpha) - \frac{\lambda}{2} \sigma_p^2(\alpha).$$

First-order condition (FOC):

$$0 = (\mu_1 - \mu_2) - \lambda \left[\alpha(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) + (\sigma_{12} - \sigma_2^2) \right].$$

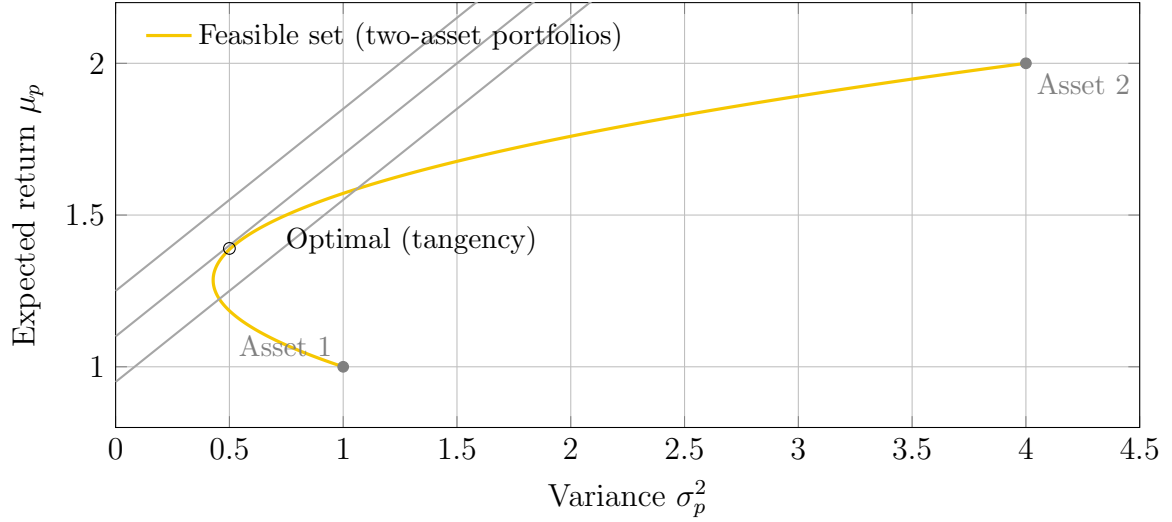


Figure 8: Mean–variance feasible set (yellow), indifference curves (gray), and optimal portfolio at tangency. The feasible set acts like a “budget constraint” for risk–return choices; preferences select the tangency portfolio.

Solve for the optimal (unconstrained) weight:

$$\alpha^* = \frac{\frac{\mu_1 - \mu_2}{\lambda} + (\sigma_2^2 - \sigma_{12})}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

Note that when $\lambda \rightarrow 0$, then the investor does not care about risk, and α^* goes to plus or minus infinity. If shorting is disallowed, this means investing everything in an asset with a higher expected return. Similarly, when $\lambda \rightarrow \infty$, then the investor cares only about risk, and then we also obtain the solution we obtained earlier.

We can illustrate the solution to the problem graphically. Figure 8 shows (i) the yellow mean–variance curve of all portfolios formed from two assets; (ii) several gray indifference lines; and (iii) the tangency (optimal) portfolio for a given λ .

6 Practice problems

6.1 Choosing a Lottery under Mean–Variance Preferences

In class, we examined how to choose between two competing lotteries. Lottery A involves rolling a fair six-sided die, where the payoff is equal to the value shown on the die in dollars. Lottery B involves tossing a fair coin six times, where the payoff is equal to the total number of heads. We previously noted that one could simply compare expected payoffs to determine the preferred lottery. Here, we extend this example to incorporate risk into the decision-making process.

Suppose Alice is risk-averse and evaluates lotteries according to a mean–variance utility function:

$$U(R) = E[R] - \frac{\lambda}{2}\text{Var}(R), \quad \lambda \geq 0,$$

where R is the payment from a lottery and λ represents her degree of risk aversion. Under this specification, Alice prefers a higher expected payoff but dislikes higher variance, which serves as a proxy for the risk associated with the lottery.

A risk-neutral Alice is characterized by $\lambda = 0$, while larger values of λ indicate a stronger marginal dislike of variability.

1. Construct a graph with the expected payoff $E[R]$ and the variance $\text{Var}(R)$ on the two axes. Plot the positions of Lottery A and Lottery B on this graph. On the same axes, sketch Alice's indifference curves and indicate the direction of increasing utility. Provide an economic interpretation of the slope of these indifference curves.
2. Determine the range of values for the parameter λ such that Alice strictly prefers Lottery A to Lottery B. Discuss the intuition behind your results.

6.2 Posted-price mechanism I

Pedro is selling his car using a posted-price mechanism. There are two potential buyers, Alice and Bob. Alice's and Bob's values are independent random variables given by, correspondingly,

$$v_A \sim U[0, 3], v_B \sim U[0, 1].$$

Buyers know their own values, while the seller does not. Pedro posts a price $p \geq 0$, and then sells to a buyer who wants to buy given this price (if more than a single buyer wants to buy then one of them is randomly chosen). A buyer $i \in \{A, B\}$ wants to buy if $p \leq v_i$.

1. What is the probability that Alice's value is higher than 1?

2. Find the variance of Alice's value.
3. What is the probability that Bob's value is higher than Alice's value?
4. Given that Bob's value is higher than Alice's, what is the probability that Alice's value is less than 0.5?
5. Let $M = \min\{v_A, v_B\}$. Find $P(M < 0.5)$.
6. Let $M = \min\{v_A, v_B\}$. Find $\mathbb{E}(M)$.
7. Suppose Pedro sets $p = 0.5$. What is the probability that at least one of the buyers will want to buy?
8. Suppose Pedro sets $p = 0.5$. What is his expected revenue?
9. Find price p that maximizes Pedro's expected revenue.

6.3 Posted-price mechanism II

Alice is selling her car using a posted-price mechanism. There are n potential buyers. Buyers have valuations that are i.i.d. with continuous uniform distribution on $[5, 10]$. Buyers know their own valuations, while Alice does not. Alice posts a price, and then each buyer decides to buy or not given this price (if more than a single buyer wants to buy then one of them is randomly chosen). Find a price (in terms of n) that maximizes the expected revenue of Alice.

6.4 Posted-price mechanism III

Alice is selling her car using a posted-price mechanism. There are 3 potential buyers. Buyers have valuations that are independently and identically distributed with the following probability density function:

$$f(x) = \begin{cases} 3x^2 & \text{if } x \in [0, 1], \\ 0 & \text{otherwise.} \end{cases}$$

Buyers know their own valuations, while Alice does not. Alice posts a price, and then each buyer decides whether to buy or not, given this price. If more than one buyer wants to buy, one of them is randomly chosen. Find a price that maximizes the expected revenue of Alice.

6.5 Revenue in selling mechanisms

Alice is selling her car. Bob and Charlie are potential buyers. Their valuations, v_B and v_C , are independent random variables drawn from a uniform distribution on $[0, 2]$. Each buyer knows their own valuation, but Alice does not know them. She is considering three different mechanisms to sell her car.

1. Posted Price Mechanism.

Alice posts a single price p . Each buyer independently decides whether to buy at that price. If both buyers want to buy, one of them is chosen at random to purchase the car.

- (a) Express Alice's expected revenue as a function of p .
- (b) Find the value of p that maximizes Alice's expected revenue. Find the optimal revenue.

2. First-Price Sealed-Bid Auction.

Each buyer i submits a bid b_i . Suppose each buyer bids $b_i = \frac{v_i}{2}$. Alice sells the car to the highest bidder, who pays the amount of their own bid.

- (a) Let X be the random variable denoting Alice's revenue. Compute $\Pr(X \leq 1/3)$.
- (b) Compute $\mathbb{E}[X]$.

3. Second-Price Sealed-Bid Auction.

Each buyer i submits a bid b_i . Suppose each buyer bids truthfully, i.e., $b_i = v_i$. Alice sells the car to the highest bidder, who then pays the second-highest bid.

- (a) Let Y be the random variable denoting Alice's revenue. Compute $\Pr(Y \leq 1/3)$.
- (b) Compute $\mathbb{E}[Y]$.

6.6 Risk and Revenue in Selling Mechanisms

Alice plans to sell her only concert ticket to one of her classmates, Bob or Carol. Both students may value the ticket differently, and Alice is unsure how much each is willing to pay. Their valuations are independent and uniformly distributed:

$$V_B, V_C \sim U[0, 2].$$

Alice herself has no value for the ticket.

Different selling mechanisms can generate different average revenues for the seller. However, Alice may also care about how *risky* her revenue is. If she urgently needs money for rent tomorrow, a smaller but more predictable revenue may be preferable to a riskier one with a higher mean. To study this, we compare the expected revenue and its variance under three mechanisms.

Second-Price Auction (SPA)

Bob and Carol each submit sealed bids equal to their valuations. The ticket goes to the higher bidder, who pays the second-highest bid.

- (a) Express the seller's revenue R_{SPA} in terms of V_B and V_C .
- (b) Compute $E[R_{SPA}]$ and $Var(R_{SPA})$.

First-Price Auction (FPA)

Each buyer bids $b_i = \frac{1}{2}v_i$. The highest bidder wins and pays their own bid.

- (a) Express the seller's revenue R_{FPA} in terms of V_B and V_C .
- (b) Derive $E[R_{FPA}]$ and $Var(R_{FPA})$.

Posted Price Mechanism (PPM)

Alice posts a price $p \in [0, 2]$. Each buyer decides whether to buy at that price. If both are willing to buy, one of them is chosen uniformly at random to purchase the ticket at price p ; otherwise, no sale occurs.

- (a) Derive the probability of sale $\Pr(\text{sale at } p)$ as a function of p .
- (b) Express the seller's expected revenue $E[R_{PP}(p)]$ as a function of p .
- (c) Find the price p^* that maximizes expected revenue and compute $E[R_{PP}(p^*)]$.
- (d) Derive $Var(R_{PP}(p))$ as a function of p and evaluate it at p^* .
- (e) Sketch $Var(R_{PP}(p))$ as a function of p . Explain intuitively why the variance approaches zero for very small or very large p .
- (f) Differentiate $Var(R_{PP}(p))$ to find the value of p that maximizes the variance. Explain why this price differs from p^* .

(g) Plot $E[R_{PP}(p)]$ on the vertical axis and $Var(R_{PP}(p))$ on the horizontal axis for all $p \in [0, 2]$. This curve represents all possible combinations of mean and variance Alice can achieve by choosing different posted prices.

- Can some prices be ruled out as *dominated* (yielding both lower mean and higher variance than another price)?
- Graphically identify the *efficient set* of prices—those not dominated by any others.
- Compare this set to the concept of the *efficient frontier* in portfolio theory.

Alice's Mean–Variance Preferences

Suppose Alice is risk-averse and evaluates mechanisms according to the mean–variance utility function:

$$U = E[R] - \frac{\lambda}{2} Var(R), \quad \lambda \geq 0,$$

where λ measures her degree of risk aversion. A risk-neutral Alice has $\lambda = 0$; larger values of λ indicate a stronger dislike of variability.

- (a) Suppose $\lambda > 0$. Compute U_{SPA} and U_{FPA} . Which mechanism does Alice prefer, and why?
- (b) Using your earlier results, write Alice's utility as a function of the posted price:

$$U(p) = E[R_{PP}(p)] - \frac{\lambda}{2} Var(R_{PP}(p)).$$

- (c) For each $\lambda \geq 0$, determine the utility-maximizing price $p(\lambda)$. Explain intuitively why a risk-averse seller might choose a price lower than the expected revenue–maximizing one.
- (d) Depict all attainable mean–variance pairs for posted prices (the frontier) together with the two auction points (SPA and FPA). On the same graph, draw several indifference curves for different values of λ . Indicate which mechanism (and which price) Alice would choose for low, moderate, and high risk aversion.

6.7 Monopolist with Uncertain Demand

A monopolist operates in a market with uncertain demand. The demand function is given by:

$$q(p) = \max\{1 - \theta p, 0\},$$

where p is the price, and θ is a random variable representing a demand shock. The CDF of θ is:

$$F(\theta) = \begin{cases} 0, & \text{if } \theta \leq 0, \\ \theta^3, & \text{if } 0 < \theta < 1, \\ 1, & \text{if } \theta \geq 1. \end{cases}$$

1. Find the median value of the demand shock θ .
2. Compute the expected value of the demand shock θ .
3. Find the probability that the revenue is strictly positive when $p = \frac{3}{2}$.
4. Find the price p that minimizes the variance of the seller's revenue.
5. Calculate the expected revenue when $p = \frac{3}{2}$.
6. Derive the expected revenue-maximizing price.

6.8 Newsvendor model I

A coffee shop owner orders fresh milk and pastry every day. Let $s_m \geq 0$ be the amount of milk ordered (in liters), and $s_p \geq 0$ be the amount of pastry ordered (in kilograms). The daily demand for milk (in liters) is a random variable $X \sim U[0, 10]$. The daily demand for pastry (in kilograms) is a random variable Y with the pdf:

$$f(y) = \begin{cases} 0, & \text{if } y \notin [0, 1], \\ 6y, & \text{if } y \in [0, \frac{1}{3}), \\ 3(1 - y), & \text{if } y \in [\frac{1}{3}, 1]. \end{cases}$$

Both X and Y are independent. The wholesale price for milk is \$1 per liter, and the shop sells milk at \$5 per liter (so the profit per liter sold is \$4). The wholesale price for pastry is \$5 per kilogram, and the shop sells it at \$6 per kilogram (so the profit per kilogram sold is \$1).

1. Find the expected daily demand for **pastry**.
2. Find the variance of the daily demand for **milk**.
3. Suppose the coffee shop orders 10 liters of milk and 1 kilogram of pastry. What is the total expected profit?
4. Suppose the coffee shop orders $\frac{1}{2}$ kilogram of pastry. Find the probability that it runs out of pastry during the day.

5. Suppose the coffee shop orders 5 liters of milk. What is the expected profit from milk sales?
6. Determine the amount of milk the shop should order to maximize its expected profit from milk sales.
7. Assume the shop owner orders the amounts of milk and pastry that maximize total expected profit. Which of the two (milk or pastry) is more likely to run out during the day? Briefly explain (no detailed calculations needed).

6.9 Newsvendor model II

Chaplin movie theater has to schedule the premiere showing of a new movie. They decide to screen the movie in two versions, original and dubbed. Let X be a random variable representing the number of viewers who come to see the dubbed version, and Y be a random variable representing the number of viewers who come to see the original version. For simplicity, assume that X is a continuous random variable uniformly distributed on $(0, 300)$; Y is a continuous random variable uniformly distributed on $(0, 200)$; and X and Y are independent. So, the dubbed version is more popular. Assume that viewers who came to see a particular version are not going to watch the other version if their preferred choice is not available.

- Suppose the cinema allocates 200 seats to the dubbed version, and 100 seats to the original.
 1. What is the expected number of viewers who come to see the movie (any version)?
 2. What is the expected number of tickets sold for the original version and for the dubbed version? Why is this number different from the previous answer?
 3. Let $Z = \max\{X, Y\}$.
 - (a) Find CDF of Z .
 - (b) Find PDF OF Z .
 - (c) Find $E[Z]$.
- Suppose now that the cinema can decide how to allocate its 300 seats across the two movies. Also let the ticket price be the same for the both movies. Find the allocation of seats that maximizes the expected revenue of the cinema. For simplicity assume that any continuous allocation is possible, that is the cinema chooses $x \in [0, 300]$ to allocate to the original, and $300-x$ to the dubbed version.

6.10 Newsvendor model III

A gas station sells gas for \$1 per liter and buys it using the wholesale price of \$0.5 per liter. A storage capacity of a gas station is 700 liters per day, that is it can store any quantity of gasoline up to this capacity. A gas station cannot sell more gasoline per day than it currently stores. Suppose the demand for gasoline is uniformly distributed between 0 and 1500. Profit of a gas station is given by its revenue from selling gasoline minus the cost of buying it from a wholesaler. Provide a numerical answer to each question.

1. Find the expected daily profit of the gas station when it decides to store 500 liters of gasoline.
2. Given the current storage capacity, find the quantity of gasoline the gas station must store in order to maximize its expected daily profit.

6.11 Newsvendor model IV

A fish restaurant sells fish for a fixed price of \$50 per kg and buys it from a supplier for a fixed cost of \$10 per kg. All fish not sold during the day goes bad. A restaurant manager has estimated that daily demand for fish in kg is uniformly distributed on $[0, 40]$. Find a profit-maximizing quantity of fish that the manger should purchase from the supplier.

6.12 Portfolio choice I

	$X = 0$	$X = 1$	$X = 2$
$Y = 0$	0.25	0.25	0
$Y = 1$	0	0.25	0.25

There are two assets. Let X and Y be random variables equal to the return on the first asset and the second asset, respectively. The joint distribution of X and Y is given by the table below.

Provide a numerical answer to each question.

1. Find the probability that X is equal to 1.
2. Are X and Y independent?
3. Suppose you know that Y is 0. Find the probability that X is also 0.
4. Find the expected value of X .

5. Find the expected value of $|X - Y|$.
6. Find the variance of X .
7. Find the covariance between X and Y .
8. Let $\alpha \in [0, 1]$ be the share of the first asset in a portfolio.
 - (a) Find α that maximizes the expected return of the portfolio.
 - (b) Find α that minimizes the variance of return of the portfolio.
9. Consider a mean-variance investor with $\lambda = 2$. Find an optimal portfolio.

6.13 Portfolio choice II

Consider two assets, whose returns are represented by random variables X and Y , respectively. Their joint distribution is provided by the following table:

	$Y = 0$	$Y = 10$
$X = 0$	0.5	a
$X = 15$	$0.5 - a$	0

Provide complete solutions and numerical answers to the following questions.

1. Assume $a = 0.3$.
 - Find the probability that $X = 0$.
 - Find the probability that $X + Y \leq 10$.
 - Find the probability that $X < Y$.
 - Are X and Y independent? Provide proof.
 - Compute the expected values $E[X]$ and $E[Y]$.
 - Compute the variances $\text{Var}(X)$ and $\text{Var}(Y)$.
 - Given that $Y = 0$, find $P(X = 0 \mid Y = 0)$.
 - Compute the covariance and correlation between X and Y .
 - Let $\alpha \in [0, 1]$ represent the share of the first asset in a portfolio. Determine the value of α that maximizes the portfolio's return.
 - Plot the portfolio return variance as a function of α .
 - Find the value of α that minimizes the portfolio return variance.

- Suppose an investor has the utility function:

$$U(\mu, \sigma^2) = \mu - \frac{1}{2}\sigma^2,$$

where μ is the expected portfolio return, σ^2 is the variance. Identify the portfolio the investor will select.

2. Now assume you can choose $a \in [0, 0.5]$.

- Find the value of a that allows you to construct a portfolio with the highest possible expected return.
- Find the value of a that allows you to construct a portfolio with the lowest variance.